

Integrated management of *Tuta absoluta* in tomato cultivation in southern Angola: a sustainable approach

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Research Article

Keywords: *Tuta absoluta*, *Solanum lycopersicum*, *Beauveria bassiana*, biological control, sustainable agriculture

Posted Date: November 3rd, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-7965828/v1>

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Additional Declarations: The authors declare no competing interests.

Abstract

The tomato borer (*Tuta absoluta*) is a highly destructive pest of tomato crops, causing losses exceeding 80% due to larval mining in leaves and fruits, which compromises photosynthesis and commercial quality. This study evaluated the effectiveness of three management strategies in family farming systems in southern Angola: chemical control, a combined system (chemical + traps), and Integrated Pest Management (IPM). The field research was conducted between 2017 and 2018, with weekly monitoring of pest population, crop damage, and productivity. The results indicated that IPM – integrating *Beauveria bassiana*, pheromone traps, and crop rotation – reduced the pest population by 86% and doubled productivity compared with chemical control. The findings demonstrate that IPM is a viable, sustainable, and economically accessible alternative for smallholder farmers, particularly under humid conditions. The study highlights the importance of promoting public policies that support the adoption of IPM and biological control agents to reduce pesticide dependence and strengthen sustainable agriculture in sub-Saharan Africa.

INTRODUCTION

In Angola, the tomato culture plays a strategic role both in food security and in generating income for family farmers. The country has a high per capita consumption of vegetables, with tomatoes being one of the most cultivated, especially in the southern provinces, such as Huíla and Namibe. However, the production chain faces structural challenges, including limited access to inputs, poor technical assistance, and dependence on imported technologies.

Tuta absoluta (Meyrick, 1917), known as the tomato leaf miner, is a devastating invasive pest of solanaceous plants, capable of causing losses of up to 100% in tomato production under unfavorable conditions. In the last five years, integrated pest management (IPM) has emerged as an effective and sustainable approach to control this pest, combining biological control, biopesticides, monitoring, and cultural management.

The proper management of agricultural pests is essential to minimize the use of agrochemicals and their negative effects on pollinators, ensuring the conservation of bees and the ecosystem services they provide. Sustainable control strategies can balance agricultural productivity with the protection of biodiversity, highlighting the importance of agricultural practices that are more harmonious with the environment (Alberto Luiz Marsaro Júnior et al., 2023).

In this context, several recent studies provide important contributions to IPM strategies in different regions: Paiva Silva et al. (2023) evaluated the repellent effects of citrus essential oils on the survival and demographics of *Tuta absoluta*, demonstrating their potential for inclusion in IPM systems (de Paiva Silva et al., 2023), Walia et al. (2024) conducted field experiments in India comparing various management modules - including chemical and biological - and identified the most effective one in reducing *T. absoluta* infestation and improving seed productivity and quality (Walia et al., 2024), Silva et al. (2021) investigated Brazilian populations of *T. absoluta* with high resistance to growth inhibitors,

highlighting the urgent need for tactical diversification within IPM to avoid failures in chemical control (J. E. Silva et al., 2021).

THEORETICAL FRAMEWORK

The theoretical framework in a study consists of a critical and organized analysis of the relevant literature on the subject, providing a theoretical contextualization and defining key concepts. It should comprehensively include the theories, models, and previous research, identifying gaps, contradictions, and consensuses in the literature that are important for the focus of the work being developed. The *Tuta absoluta* (Meyrick, 1917) is one of the most destructive pests of tomato crops, potentially causing losses greater than 60% when not managed properly (Mohamed et al., 2022). Research on its biology indicates that the complete cycle from egg to adult ranges from 19 to 54 days, allowing for up to 13 annual generations in tropical regions with temperatures between 15°C and 30°C.

In biological control, the parasitoid *Trichogramma pretiosum* exhibits a type II functional response and can parasitize up to approximately 28% of the pest's eggs, as demonstrated in experimental studies (Dalbianco et al., 2025). Recent research with different strains of this agent, such as the S2 strain, has demonstrated greater efficiency on populations of *T. absoluta* collected in 2022 and 2023, indicating potential for practical application in Integrated Pest Management (IPM) programs. The entomopathogenic fungus *Beauveria bassiana* has also stood out as a viable alternative, achieving larval mortalities greater than 90% under laboratory conditions. Such results were observed both with the application of spores and with toxins extracted from the BMAUM-M6004 strain (A. C. L. Silva et al., 2020).

Furthermore, tomato plants endophytically inoculated with *B. bassiana* showed a reduction in leaf consumption and an increase in larval mortality of *T. absoluta* through ingestion or contact (Khaled et al., 2023). Natural enemies like *Trichogramma pretiosum* parasitize eggs, with effectiveness between 40–60% (Fagundes et al., 2024). Entomopathogenic fungi (*Beauveria bassiana*) cause larval mortality of up to 85% under high humidity conditions (Chouikhi et al., 2022).

Alternative host plants include eggplant (*Solanum melongena*) and potato (*Solanum tuberosum*), which support the movement between crops (J. E. Silva et al., 2016). Populations have developed resistance to pyrethroids (lambda-cyhalothrin) and spinosads (spinosad) due to continuous use (Li et al., 2018). Mutations in the RyR (ryanodine) gene confer resistance to diamides (Douris et al., 2017b, 2017). The rotation of modes of action (IRAC Groups 5, 6, and 28) is essential for sustainable management (Nauen et al., 2019). A detailed understanding of the biological developments of *Tuta absoluta* is essential for the development of effective control strategies.

In this context, it becomes feasible to identify the critical phases of pest development, which allows for the targeting of more effective control strategies, such as the application of insecticides at the moment of egg hatching or the use of specific traps to capture adults.

As mentioned earlier, the eggs are laid individually on leaves (Ebrahimi et al., 2019), larvae cause damage directly by mining plant tissues, while the pupae develop in protective cocoons (Bruwer et al., 2021a), and the adults are responsible for the spread of the pest (Alkemade et al., 2022). This visual representation complements the cited data, highlighting the need for specific specifications for each phase of the biological cycle, as detailed in the specifications above. In light of the challenges posed by *Tuta absoluta* to tomato production in southern Angola, this study aimed to compare the agronomic efficacy and economic viability of three different pest management strategies: conventional chemical control, a combined system (chemical + homemade traps), and Integrated Pest Management (IPM), under real smallholder farming conditions, aiming to identify sustainable practices suited to the regional agroecological context.

MATERIAL AND METHODS

The study was conducted between 2017 and 2018 on three family properties located in the provinces of Huíla and Namibe, in southern Angola, representative of the regional production of tomatoes. The Quavi farm is located in Lubango (14°54'32"S, 13°30'15"E, 1,763 m above sea level); the Joazivino farm, in the municipality of Tombwa (15°10'12"S, 12°09'48"E, 512 m); and the Muhambo farm, in the municipality of Moçâmedes (15°11'30"S, 12°08'00"E, 519 m). These coordinates correspond to the experimental areas used for comparison between different management systems of *Tuta absoluta*.

The experimental design adopted two repetitions per locality, totaling six experimental units distributed over areas of 400 m² (Joazivino), 417.5 m² (Muhambo), and 409.5 m² (Quavi) per repetition, totaling 2,454 m² of experimental area. The small dimensional variations (< 5% between plots) were compensated by standardizing the data in kg/m² and percentages to ensure comparability. Each repetition received a standardized density of 20 traps per hectare and 20 monitored plants, with weekly assessments of the *Tuta absoluta* population and damage to the crop.

Three management systems were compared: Conventional chemical control (Joazivino), using exclusively the insecticides Suntap 500 SP (50 g/100 L) and Karate 5EC (1 kg/100 L) applied every 15 days during the times of highest pest activity (6 a.m. – 8 a.m. and 3 p.m. – 5 p.m.); the intermediate system (Muhambo) that combined these same insecticides with homemade traps (yellow-painted drums with internal light and glue) and rotation with watermelon; and Integrated Pest Management - IPM (Quavi), which integrated weekly nighttime applications of *Beauveria bassiana* (strain ESALQ PL63, 2×10⁸ spores/mL), light traps and Delta traps with pheromones, tomato-lettuce rotation for 60 days, soil solarization, and selective application of abamectin (0.5 mL/L) only when infestation exceeded 5 larvae/plant.

The evaluated parameters included: Weekly population monitoring (adults captured by trap and number of larvae per plant); quantification of foliar and fruit damage through standardized visual assessment; productivity measured by the weight of marketable fruits per square meter; and daily recording of

environmental variables (relative humidity and temperature) to correlate with the effectiveness of the treatments.

The statistical analysis was performed using ANOVA in a randomized block design (considering the six repetitions as blocks), with mean comparison using Tukey's test ($p < 0.05$), whose coefficients of variation between 10.9% and 15.8% attested to the methodological robustness against natural variations among properties.

As methodological limitations, we highlight: (1) the operational differences between properties, which prevented the absolute standardization of the plots, although compensated by statistical analysis and metric standardization of the data; (2) the sensitivity of biological control to climatic conditions, with the effectiveness of *B. bassiana* varying from 68% larval mortality at relative humidity $> 70\%$ to 45% in drier conditions ($RH < 60\%$); and (3) the assessment period limited to two agricultural cycles, which, while sufficient to demonstrate trends, requires multi-year studies to evaluate long-term effects on pest population dynamics. These limitations, inherent to studies in real family farming conditions, do not compromise the conclusions but highlight opportunities for future research with a larger number of properties and combinations with other biological control agents.

In chemical control, the insecticides Suntap 500 SP (50 g/100 L; emamectin benzoate) and Karate 5EC (1 kg/100 L; lambda-cyhalothrin) were applied every 15 days, following optimized protocols for the management of *Tuta absoluta*. The applications were carried out at two strategic times: (i) between 6 a.m. and 8 a.m., a period of favorable environmental conditions (low wind speed and less intense solar radiation), which minimizes drift and enhances product deposition on the foliage; and (ii) between 3 p.m. and 5 p.m., coinciding with the peak of the pest's crepuscular activity (adult flights and larval movement), as described in the literature for Lepidoptera (Altesor & González, 2023; Zeng et al., 2024). This approach aims to maximize the contact of contact/ingestion insecticides with the biological target, respecting the ecophysiological characteristics of the crop and the pest.

This study was conducted in real field conditions, focusing on practical comparisons between different management strategies. Due to logistical and operational limitations of the participating properties, it was not possible to apply a completely randomized experimental design with blocks and standardized repetitions. To mitigate this factor, systematic monitoring by farm was adopted, with regular records and control of climatic and operational variables. This approach enabled the collection of representative and useful data for the context of family farming in southern Angola.

The data regarding the population density of *Tuta absoluta*, the incidence of damage, and productivity were organized into spreadsheets and subjected to statistical analysis through analysis of variance (ANOVA), using the R software. When significant differences were identified ($p < 0.05$), the means were compared using the Tukey test. This procedure allowed for the evaluation of the effectiveness of the treatments (chemical control, combined system, and ICM) in reducing the pest population and increasing the productivity of tomato plants.

RESULTS AND DISCUSSIONS

The present study evaluated the effectiveness of different strategies for controlling *Tuta absoluta* in tomato crops in southern Angola, focusing on economically viable methods for small producers. The evaluation was conducted on three farms, representative of the edaphoclimatic conditions of the region, using distinct management systems: Conventional chemical control (Joazivino), chemical control associated with homemade traps (Muhambo), and Integrated Pest Management - IPM (Quavi).

Considering the monitoring carried out on 20 plants per farm, the data presented in (Table 1) reveal marked differences in the effectiveness of the methods employed, as well as the number of adults captured, the efficiency of the traps, and the impacts on damage and productivity. The system based exclusively on chemical control (Joazivino) showed the worst performance, with a population reduction of only 41% after eight weeks. On the other hand, the strategy that combined chemical control with homemade traps (Muhambo) resulted in a 63% reduction.

Table 1

Average number of *Tuta absoluta* adults captured and percentage of efficiency by treatment in experimental field. Namibe, Angola - Mandume Ya Ndemufayo University (*Universidade de Mandume Ya Ndemufayo* - UMN), in collaboration with USP/ESALQ, 2018.

Treatment / Location	Management Strategy	Mean number of adults captured (per trap)	Reduction (%)	Efficiency (%)
Joazivino (Namibe)	Chemical control (Suntap + Karate 5EC)	18.7 ± 3.2	—	41%
Muhambo (Namibe)	Chemical + Homemade traps + Crop rotation	10.8 ± 2.5	63%	63%
Quavi (Huíla)	Integrated Pest Management (IPM) – <i>B. bassiana</i> + Pheromone and light traps + Crop rotation	2.6 ± 0.9	86%	86%

Source: Field data (2017–2018).

Means followed by the same letter in the column do not differ significantly (Tukey, $p < 0.05$).

As seen in Table 2, the impacts of different treatments on leaf and fruit damage, as well as on productivity. At Quavi farm (IPM), only 18.5% of leaves were damaged and 12.4% of fruits were affected, while at Joazivino (chemical control), the values were 65.3% and 48.7%, respectively.

Productivity also reflected these differences: Quavi achieved 5.2 kg/m², more than double the productivity recorded at Joazivino (2.1 kg/m²). The Muhambo farm showed intermediate results: 39.1% of damaged leaves, 28.6% of affected fruits, and productivity of 3.7 kg/m² (Table 2). These data reveal a direct relationship between the intensity of damage and final productivity. There is an inverse correlation between the percentage of damaged leaves and productivity: As damage increases, yield decreases.

Table 2

Control efficiency, crop damage, and productivity in three management systems of *Tuta absoluta* in Southern Angola (2017–2018).

Farm / Location	Management System	Damaged Leaves (%)	Damaged Fruits (%)	Population Control Efficiency (%)	Productivity (kg m ⁻²)	Cost (USD kg ⁻¹)
Joazivino (Namibe)	Chemical Control	65.3 ± 4.1	48.7 ± 3.5	41	2.1 ± 0.3	0.42
Muhambo (Namibe)	Chemical + Traps	39.1 ± 3.8	28.6 ± 2.9	63	3.7 ± 0.4	0.28
Quavi (Huíla)	Integrated Pest Management (IPM)	18.5 ± 2.6	12.4 ± 1.8	86	5.2 ± 0.5	0.19

Source: Field data (2017–2018).

Means followed by the same letter in the column do not differ significantly (Tukey, $p < 0.05$). Data collected at Quavi (Huíla), Joazivino, and Muhambo (Namibe) farms.

The three management systems differed in all the evaluated parameters: population control, visual damage, and economic return. The Integrated Pest Management (IPM) stood out with the best indices in all variables: 86% efficiency in population control, the lowest level of damage (12.4% in fruits), highest productivity (5.2 kg/m²), and the lowest cost per kilogram produced (0.19 USD/kg) (Tables 1 and 2).

As seen in (Table 1) and Fig. 1, the variation of the *Tuta absoluta* population over time. At the Quavi farm, the population dropped drastically starting from the third week, remaining below the economic damage threshold until the end of the cycle. In Joazivino, on the other hand, continuous growth was observed, with a peak of 18.7 ± 3.2 individuals per trap in the fourth week - a value 3.7 times higher than the damage threshold. The Muhambo farm showed an intermediate behavior, with a progressive reduction of the population starting from the fourth week.

The second-degree polynomial regression for Quavi ($R^2 = 0.94$) indicated effective and sustainable control of the pest population, while the equation for Joazivino ($R^2 = 0.91$) reflected uninterrupted growth. These results reinforce the importance of the combined use of traps, biological control, and cultural strategies, such as crop rotation, to rapidly reduce the infestation of *T. absoluta* and keep it below the economic damage threshold (Table 1, Fig. 1).

The visual impacts and the main inputs of the management are highlighted in Fig. 2, with emphasis on the artisanal traps, responsible for 38% of the adult captures in Quavi. The technical limitations of conventional spraying were also evidenced, while the risks of applications without PPE were highlighted. The leaves and fruits showed typical symptoms of pest attacks. These evidences confirm the technical

and practical superiority of IPM, especially in contexts of family farming with limited access to commercial technologies.

The success of the IPM heavily depends on favorable environmental conditions. The effectiveness of *B. bassiana* is directly related to humidity, and soil solarization, crop rotation, and elimination of cultural residues have proven essential to reduce pupa populations. It is also crucial to ensure that producers have access to bioinputs and ongoing technical training.

The excellent model fit ($R^2 = 0.94$) proved the effectiveness of this approach, which combined biological control (*Beauveria bassiana*), light traps, and crop rotation. The Intermediate System (Muhambo): Achieved better results than the isolated chemical, but lower than the IPM, with population reduction only starting from the 4th week.

These visual results corroborate the quantitative data from the study, reinforcing that the IPM not only controls the pest more effectively but also keeps its population at safe levels over time, unlike conventional methods that allow uncontrolled infestations. The figure highlights the importance of an integrated approach for sustainable agriculture in the region.

These results reinforce that the effectiveness of traps in managing *T. absoluta* is intrinsically linked to the materials and methods used in their manufacturing. The success of traps for *T. absoluta* depends on both the attractiveness of the pheromones and the design and location of the traps, according to (Plata-Rueda et al., 2020).

As shown in Fig. 2, the combination of low-cost and high-efficiency materials follows the principles of Integrated Pest Management (IPM), advocated by (Desneux et al., 2010), as an essential approach to reducing dependence on chemical insecticides. The use of handcrafted traps, when properly sized, can yield results comparable to commercial methods, as demonstrated by Plata-Rueda et al. (2020) in studies on mass capture of *T. absoluta*.

This practice, besides being economically viable, is aligned with the demands for sustainability in agriculture, an aspect highlighted by Van Middelhaar et al. (2013) in analyses on agroecological innovations. This comparison reinforces the need to integrate chemical methods with other strategies of Integrated Pest Management (IPM), such as biological and physical control, to ensure sustainable effectiveness and reduce environmental and health risks.

The low use of chemical insecticides (only abamectin in critical situations) and the high efficacy of *B. bassiana* (68% mortality) support studies like those of Chouikhi et al., (2022), that highlight the potential of entomopathogenic fungi in tropical climates.

At the same time, the use of backpack sprayers has shown technical problems in the uniform distribution of pesticides, compromising their efficiency. These findings reinforce the need to integrate chemical methods with other IPM strategies to ensure sustainable and effective pest control. The commercial product, containing [active ingredient, if known, e.g.: "cypermethrin (0.15%)"], represents the

conventional control approach, whose effectiveness has been questioned due to the development of resistance, as warned by (W. M. Silva et al., 2015).

The comparative analysis highlighted critical differences in the practices of applying agricultural pesticides. While the correct use of PPE ensured safety for the applicator and alignment with pest management regulations, spraying without protection resulted in elevated risks of human and environmental contamination, as well as enhancing pest resistance. The adoption of safe practices and access to protective equipment remain challenges among family farmers, which requires investments in rural extension and continuous training. The damage caused by *T. absoluta* was severe, both on the leaves and the fruits. On the leaves, extensive necrosis and the formation of galleries by mining larvae were observed, drastically reducing the photosynthetic area.

These findings corroborate studies such as those by Yan et al. (2020), which highlight the viability of IPM in tropical regions, especially when combined with biological control and cultural practices.

The effectiveness of *B. bassiana* (68% larval mortality) was influenced by humidity, with better results in high relative humidity conditions (> 70%). This variation had already been reported by (Chouikhi et al., 2022), highlighting the importance of environmental monitoring to optimize applications. Furthermore, studies like Marassatto's (2022) have shown that the endophytic colonization of *B. bassiana* (strain ESALQ PL63) in soybean plants can induce systemic resistance against pests like Spodoptera frugiperda, demonstrating the potential of this strain in sustainable pest management (Marassatto, 2022).

Crop rotation and soil solarization, adopted in Integrated Pest Management, reduced the survival of pupae by 45%, a strategy validated by Bruwer et al. (2021) in other contexts. Economically, the IPM proved to be viable for family farmers, with a cost of USD 0.19/kg (vs. USD 0.42/kg for chemical control). This advantage aligns with recent meta-analyses Romero Antonio et al. (2025), that associate agroecological practices with gains in productivity and sustainability.

Furthermore, the limited number of participating properties and the absence of controlled statistical delineation reduce the power of data generalization. Studies with greater geographical coverage and methodological rigor are essential to validate the effects of IPM in different agroecological contexts.

Although the results clearly demonstrate the effectiveness of the IPM, it is important to consider that the study was conducted under real field conditions, where logistical limitations prevented the adoption of a completely randomized experimental design with standardized repetitions. This methodological particularity, inherent to applied research in family farming contexts, suggests caution in extrapolating the data to other regions or large-scale production systems. Additionally, the variable effectiveness of biological control with *Beauveria bassiana* under low humidity conditions (< 60%) highlights the need to develop more stable formulations or application protocols adapted to the semi-arid climate.

Another relevant aspect is the relatively short evaluation period (two agricultural cycles), which limits the understanding of the long-term effects of the IPM on pest population dynamics and plant resistance. However, these limitations do not invalidate the conclusions, but highlight opportunities for future research, such as multi-year studies involving a larger number of properties and the combination of *B. bassiana* with other biological control agents, such as *Trichogramma pretiosum*, to enhance the results.

The obtained results are in line with the literature. Studies with *Beauveria bassiana* demonstrated mortality of *T. absoluta* larvae varying between 90–95% in endophytically colonized plants (Chouikhi et al., 2022), while commercial formulations reached 86.8% efficacy 86,8 % de eficácia (Khaled et al., 2023).

Furthermore, meta-analyses with agroecological practices reveal an average productivity increase of 39% compared to monocultures, also indicating economic benefits (Romero Antonio et al., 2025). The present study reinforces these findings by demonstrating, under real field conditions in Angola, that the IPM not only effectively controls the pest but also provides greater profitability per hectare. There is also a need for technical training for the precise application of biological agents, as well as public policies that facilitate access to these inputs for small farmers.

In this way, the data presented here provide important subsidies for rural extension programs, as well as highlighting the need to expand participatory research focused on climate adaptation and agroecological transition in semi-arid environments. Future studies could explore the combination of *B. bassiana* with other biological agents, such as *Trichogramma pretiosum*, in different regions of Angola. Furthermore, investigations into the economic viability of local bio-input production and the creation of standardized application protocols could enhance the impact of Integrated Pest Management (IPM) on family farming.

In this regard, public policies aimed at valuing family farming should prioritize technical support and ongoing training for producers on sustainable practices such as Integrated Pest Management (IPM). Partnerships between universities, research institutes, NGOs, and local cooperatives can facilitate the adoption of these technologies, reducing dependence on imported inputs and promoting greater productive autonomy in rural communities. It is emphasized that the systematic monitoring of climatic variables and the operational control adopted have partially mitigated these restrictions, ensuring representative data for the analyzed context.

This study demonstrated that Integrated Pest Management (IPM), based on biological control with *Beauveria bassiana*, light traps, and crop rotation, is the most effective strategy for reducing *Tuta absoluta* infestations in tomato plants in southern Angola, with an 86% reduction in population and an increase in productivity to 5.2 kg/m². The economic viability of the system (cost of USD 0.19/kg) reinforces its potential for family farmers, reducing the dependence on chemical insecticides.

The logistical limitations of the study, such as the lack of randomization, do not compromise the validity of the results, but highlight the need to expand research to other edaphoclimatic regions of Angola and

test combinations with other biological control agents (e.g., *Trichogramma pretiosum*). It is recommended that public policies encourage:

1. The technical training for the application of the IPM;
2. The local production of bioinputs, and

The creation of climate monitoring networks to optimize the timing of interventions. These actions can transform the findings of this work into scalable solutions aligned with the United Nations Sustainable Development Goals (SDGs).

Declarations

CLINICAL TRIAL NUMBER

Not applicable.

FUNDING

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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Figures

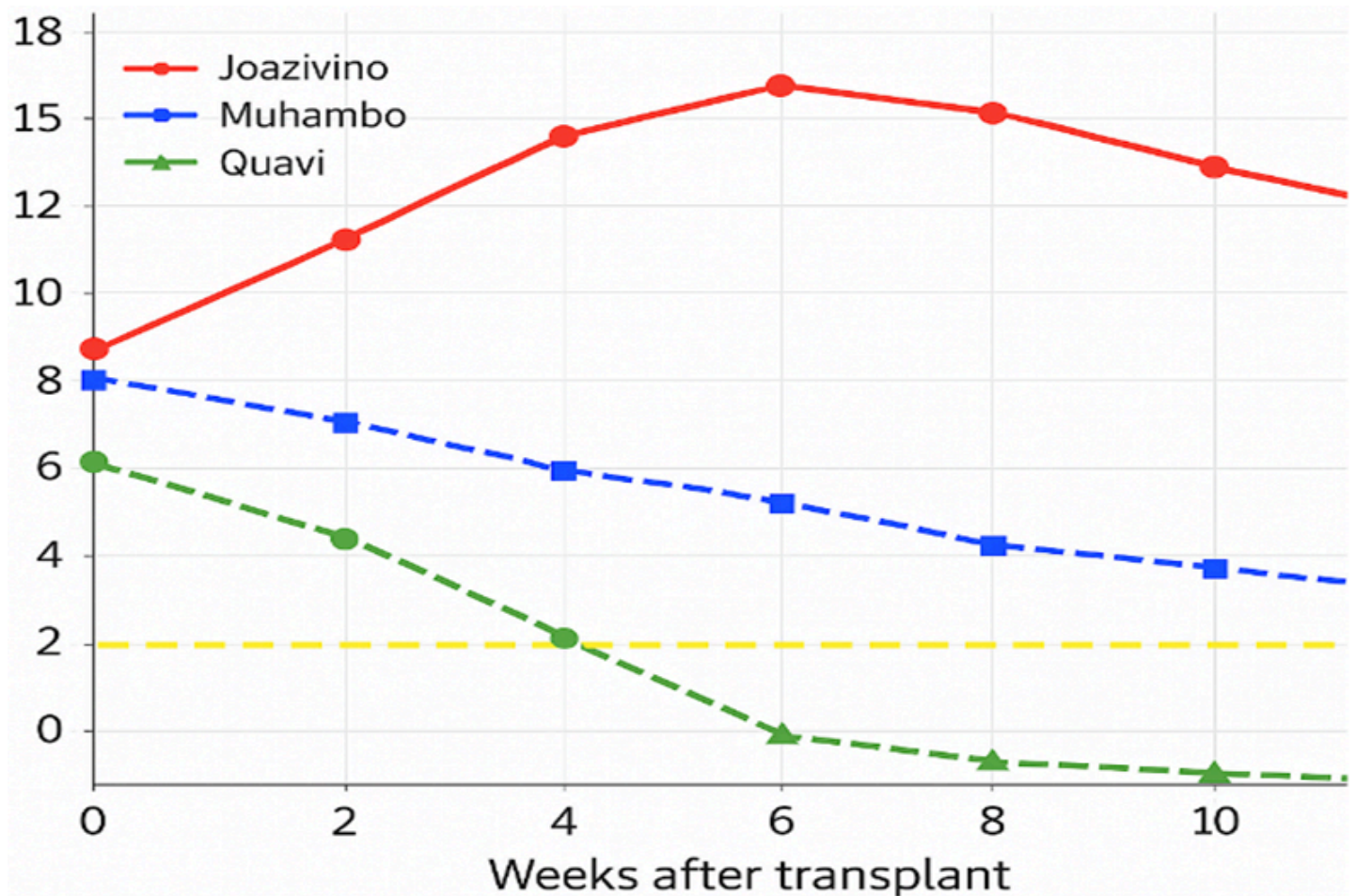


Figure 1

Population dynamics of *Tuta absoluta* over 12 weeks after transplanting in three locations. Lubango - Angola, 2018.

Source: Field data (2017–2018).



Figure 2

Handmade trap used in *Tuta absoluta* control: (A) Manual assembly of the device; (B) Field installation. Lubango - Angola, 2018.

Source: Field data (2017–2018).



Figure 4

Handmade trap used in control: (A) Manual assembly of the device; (B) Field installation. Lubango - Angola, 2018.